The Sondondo Valley Project: A Narrative of Work (June–Sept 2026)

(Prepared as a guide for resuming the project)

Part 1: What We Accomplished

1. Linkage-ready data, with the originals protected. We added event dates, places, record types, and cleaned versions of every name to the persona file. All 47,072 original mentions were kept exactly as transcribed. The cleaned names are used only for matching.

2. A first set of confirmed matches. An early, strict pass found:

- 82 duplicate mentions, merged into 69 people.
- 163 links between record types: 82 baptism-to-marriage, 52 marriage-to-burial, and 29 baptism-to-burial.

Almost all of these were confirmed by expert review.

3. A rulebook of matching rules. Historical knowledge about how people, families, and names appear in these records was turned into clear rules the computer can follow. They are described in Part 2.

4. A name-frequency analysis. This table lists every name, how often it appears, whether it is male or female, and how common it is ("Most common" to "Very rare"). It was reviewed and improved through 13 versions. Along the way:

- Names of unknown sex fell from 710 to about 70.
- 284 long, multi-part names were checked and split into name, middle name, paternal surname, and maternal surname.
- A few records that had been merged by mistake were separated.

5. A "safe release" of linked people and families. This is the main result:

Measure	Result
Original persona mentions	47,072
Mentions securely linked into identities	41,251
Mentions not yet linked to anyone else	5,821
Distinct people	44,995
Family clusters	29,013
Family clusters with more than one person	7,156

Measure	Result
Trusted family relationships (parent–child and spouse)	16,281
Errors (relationships crossing between families)	0

This includes 1,564 people, built from 3,641 mentions, who appear in more than one record. They are our most secure individuals. Every original mention can be traced to the person and family it was assigned to.

6. Working family trees. Sample trees, such as the Chinchay, Llamocca, and Quispe families, show parents, children, and spouses taken directly from the linked data. A manual check confirmed that the method correctly kept two women with the same name (Nieves Guillén) in separate families. The families can be opened in common genealogy and network tools.

7. Possible matches waiting for review. These were found but deliberately **not** merged:

- 480 possible sibling links, including 452 possible brothers and sisters.
- 8,089 possible matches for godparents at marriages. This work is paused for now.

8. Organized code and data. All work is in three notebooks, run in order, plus a numbered data package. See "Where things are kept" below.

Part 2: How We Accomplished It

Step 1: Setting the priorities

We agreed early on what counts as evidence, in this order:

1. **Sex:** a man and a woman cannot be the same person.
2. **First name and surname.**
3. **Parents' names.**
4. **Dates.**
5. **Place:** supporting evidence only. It can never create a match on its own.

Step 2: Preparing the data

Computers match text exactly. "María Guarcaya" and "Maria Huarcaia" look completely different to a computer. So we created a standardized version of each name for comparison and left the original untouched.

To avoid comparing every mention with every other one, which would mean over a billion comparisons, we only compared people who share basic features, such as the same surname and first initial. Limits were added so that very common names could not produce huge numbers of weak comparisons.

Step 3: Two methods that check each other

- **Splink** is a free record-linkage tool. It gives each possible pair a *probability* of being the same person, based on how well names, parents, and dates agree. We kept only very high-probability matches (95–99%).
- **Rules plus scoring** sorts pairs by the type of family evidence they share, for example "same father and same mother at different baptisms." A machine-learning model helps rank these cases, but it does not merge anything on its own.

Our main principle was **caution first**. Wrongly joining two different people does more harm than missing a real match.

Step 4: Expert review

Dr. de la Puente reviewed samples of the results, often against the original parish-book images. His decisions were used to correct the results and to create new rules. We repeated this cycle many times.

Step 5: The main rules

- **Impossible pairs are excluded,** such as a man matched with a woman, or two people in the same role at the same event.
- **Events must be in a possible order:** baptism, then marriage, then burial. Clearly impossible cases are rejected, and unclear ones go to review.
- **Minimum ages:** about 12 for marriage, and at least 12 for serving as a godparent or witness.
- **Burial age check:** if the age at burial is within 15 years of the age calculated from the baptism, the match is supported. A bigger difference goes to review.
- **Same spouse = strong match:** if the name, spouse, and dates agree between a marriage and a burial, it is a match even if the parents are missing.
- **Baptism as a tie-breaker:** if the parents at marriage and at burial disagree, the baptism record decides.
- **Same-named couples 50+ years apart are different families.**
- **Rare names strengthen a match;** common names need extra support from relatives or dates.
- Every accepted match gets a **provisional ID** that can be traced back to its records.

Step 6: Name cleaning and name-frequency analysis

Good matching depends on clean names, so:

- **Placeholders** that only meant "not recorded" were removed.
- **Abbreviations** were expanded, for example "Ma" to "María".
- **Missing sex** was filled in from the person's role (a godmother is female) and from a reference dataset from the Chipao parish.

- **Known spelling variants** were grouped, for matching only:
    - **Accents are ignored:** Estévan = Estevan.
    - **"ce/se" and "ci/si" count as the same sound:** Nepomuceno = Nepomuseno, Dionicio = Dionisio.
    - **34 other variants** were approved one by one, each with supporting evidence.
- **New possible variants** go on a review list and are never merged automatically.

The frequency of each name was then calculated using these cleaned names, which shows how "rare" a shared name really is.

Step 7: Building families

We grouped people into **family clusters**. Each person keeps their own ID, and verified relatives share a family ID. Only trusted relationships are included.

To find siblings, we used a method proposed by Dr. de la Puente:

1. Take a bride or groom whose parents are known.
2. Estimate their birth year.
3. Search the baptism records for other children of the same parents within about 15 years.
4. Allow for children born before the parents married.

Godparent matching was paused. A single transcription error ("Oseas" read as "Oscar") showed how easily a wrong match could slip through. When it resumes, it will only look for people already confirmed as unique adults.

Step 8: Showing family trees

The planned display is an **"hourglass chart."** One person sits in the center, with ancestors above and descendants below. The goal is 3–4 generations in each direction.

Where to Pick Up When the Project Resumes

1. Review the 480 possible sibling links.
2. Compare the 1,564 secure individuals with the family clusters.
3. Extend the family trees to grandparents and grandchildren.
4. Resume godparent matching, only for confirmed unique adults.
5. Find second and third marriages. Many "different spouse" cases turned out to be remarriages.
6. Make fuller use of place names.
7. Add a second parish (Aucará/Chipao) and recheck the name-frequency table.

8. After every rerun, compare the results with the previous version before accepting new rules.

Short Glossary

- **Mention (persona):** one appearance of a name in one record.
- **Record linkage:** deciding which mentions belong to the same real person.
- **Family cluster:** people connected by trusted parent–child or spouse links.
- **Safe release:** matches we are confident in. Everything else stays in review.